FULL STACK SOFTWARE ENGINEERING CAPSTONE PROJECT

GiftLink

A Cloud-Native Microservices Platform for Community Resource Sharing with Natural Language Sentiment Analysis

Project Title:	GiftLink - Community Gift Sharing Platform
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Technology Stack:	MERN Architecture (Node.js, Express, React 18, MongoDB, Natural NLP)
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Executive Summary

GiftLink is an innovative, cloud-native full stack web platform designed to facilitate community resource sharing, waste reduction, and peer-to-peer gifting of household goods. Grounded in principles of environmental sustainability and circular economy, GiftLink provides an intuitive and engaging environment where community members can browse available pre-loved items, post gifts, search items with multi-faceted filtering criteria, interact with fellow neighbors through community discussions, and manage authenticated profiles.

Built upon a scalable microservices architecture, GiftLink leverages React 18 and Bootstrap on the frontend, Express.js and Node.js for high-throughput REST APIs, MongoDB for flexible JSON-native document persistence, and a specialized Natural Language Processing (NLP) microservice running Natural's Porter Stemmer and AFINN sentiment analysis to evaluate community comments and ensure a wholesome, constructive exchange ecosystem.

> **CAPSTONE PROJECT VISION**
> *GiftLink demonstrates an end-to-end full-stack software engineering lifecycle, incorporating secure stateless JWT authentication, cryptographic password hashing with Bcrypt, reactive single-page application state management, decoupled REST APIs, and an AI-powered microservice for automated sentiment analysis.*

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1. Project Overview & Problem Statement

1.1 Background & Motivation

Modern urban societies generate millions of tons of solid municipal waste annually, a significant proportion of which consists of fully functional, lightly used goods such as furniture, appliances, kitchenware, books, and decor items. Traditional disposal methods result in landfill saturation, unnecessary municipal expenditure, and heightened carbon footprints caused by manufacturing replacements.

Simultaneously, countless households, students, and low-income community members actively seek cost-effective avenues to furnish living spaces or obtain essential household utilities. Although classifieds and commercial resale platforms exist, they are frequently encumbered by monetary transaction fees, shipping hurdles, scams, and aggressive commercial advertisements, which deter purely altruistic giving.

1.2 The Problem Statement

The challenges addressed by this project include:

- **Underutilization of Reusable Resources:** Households dispose of functional goods prematurely due to the absence of convenient, localized giving channels.
- **Monetization Friction & Barriers:** Mainstream marketplaces focus on commercial monetization rather than communal benevolence and zero-cost resource transfer.
- **Absence of Dedicated Community Platforms:** Dispersed online groups often lack standardized item categorization, structured search, condition tracking, and search filters.
- **Toxicity & Negative Communication:** Unmoderated online donation forums frequently suffer from toxic, disrespectful, or scam comments, degrading user trust.

1.3 The GiftLink Solution

GiftLink resolves these challenges by introducing a purpose-built, zero-cost community gifting ecosystem. By pairing a user-friendly, responsive Single Page Application (SPA) with a resilient microservices backend and an automated sentiment evaluation engine, GiftLink empowers community members to give items a second life, eliminate waste, and foster neighborhood solidarity.

2. Project Objectives & Scope

2.1 Primary Objectives

- **Architectural Modularity:** Develop a modular, scalable architecture separating client presentation, core business logic, database management, and AI NLP services.
- **Enterprise-Grade Authentication:** Implement secure sign-up, login, salted password hashing, stateless JSON Web Token (JWT) issuance, and authorization guards.

- **Intuitive Catalog Discovery:** Provide an intuitive interface to browse gifts by category, condition, date posted, and age with zero friction.
- **Multi-Dimensional Search & Filtering:** Build a multi-parameter search engine enabling text queries, dynamic category filtering, condition classification, and age threshold sliders.
- **Integrated Sentiment Intelligence:** Integrate an autonomous Natural Language Processing service to analyze user comments and score community interaction sentiments.
- **High Performance & Observability:** Deliver high-speed query execution with MongoDB and structured real-time server logging with Pino for observability.

2.2 System Scope & Stakeholders

The platform serves three core stakeholder personas:

- **Donors / Item Providers:** Individuals with surplus items who wish to donate them quickly, securely, and transparently to neighbors.
- **Recipients / Claimants:** Community members, students, and families looking for free furniture, living supplies, electronics, or books.
- **Community Moderators & Admins:** Oversees platform hygiene, analyzes comment sentiment health, and monitors API metrics and database growth.

3. System Architecture & High-Level Design

GiftLink follows a modern, decoupled microservices architectural pattern. Each functional tier maintains single responsibility, enabling horizontal scalability, isolated fault domains, and independent service deployment.

> **SYSTEM DATA FLOW & ARCHITECTURE PIPELINE**
> *Client (React SPA / Static HTML) ➔ Reverse Proxy / API Gateway ➔ Core Express Backend (Auth, Gifts, Search) ➔ MongoDB Database Cluster (giftdb) ⟷ Sentiment Analysis Microservice (Natural NLP / Express)*

3.1 Architectural Component Breakdown

Architectural Tier	Underlying Engine	Functional Responsibility
Presentation Layer (giftlink-frontend)	React 18 SPA & Bootstrap 5	Renders dynamic user interfaces, routes (App.js), Auth Context state provider, responsive product grids, and interactive filtering controls.
Static Entry Portal (Public Landing)	HTML5, CSS3, Google Fonts	Serves fast-loading landing portal (home.html) introducing the GiftLink mission, value proposition, and instant navigation into the application.
Core Backend Service (giftlink-backend)	Node.js & Express.js (Port 3060)	Orchestrates API endpoints (/api/auth, /api/gifts, /api/search), connects to MongoDB, enforces token

		auth, and logs telemetry via Pino.
Sentiment Service (sentiment)	Node.js & Natural NLP (Port 3000)	Dedicated microservice evaluating user sentiments using Porter Stemmer tokenization and AFINN dictionary sentiment calculation.
Persistence Layer (MongoDB Database)	MongoDB NoSQL ('giftdb')	Stores JSON-native document collections ('users' and 'gifts') with indexing on gift attributes, timestamps, and user credentials.

4. Functional Modules & Core Features Specification

4.1 User Authentication & Authorization Management

Security is central to GiftLink. The authentication module manages user onboarding, credential hashing, and session persistence without server-side state lock-in:

- **User Registration:** Users register with first name, last name, unique email address, and password. The system checks database uniqueness, enforces salting (10 rounds with bcryptjs), and securely inserts user records into MongoDB.
- **Secure User Login:** Registered users authenticate with their email and password. Upon successful validation, the backend generates a signed JSON Web Token (JWT) encapsulating user identity payload with secret signing keys.
- **Stateful Session Persistence:** React Context (AuthContext.js) maintains global authentication state across all components. Session tokens and user display details are maintained in browser sessionStorage, enabling seamless navigation across page refreshes.
- **Profile Customization:** Authenticated users can view their profile details, inspect registered email addresses, and update their display names dynamically via API-backed profile management.

4.2 Gift Catalog & Feed Discovery

The primary interface of GiftLink (`MainPage.js`) displays active community gift listings retrieved from `/api/gifts`. Key capabilities include:

- **Dynamic Item Cards:** Responsive card grid rendering high-resolution product thumbnails, item titles, condition badges, and human-friendly post dates.
- **Condition Class Badging:** Color-coded visual badges clearly distinguishing item states: 'New' (success green) vs. 'Like New' and 'Older' (warning amber).
- **Date Formatting:** Raw Unix epoch timestamps are converted to readable locale dates (e.g., 'August 28, 2023') for seamless user perception.
- **Product Deep Linking:** Dedicated call-to-action buttons redirecting users directly to the comprehensive product details view.

4.3 Advanced Multi-Faceted Search & Filtering Engine

The search module (`SearchPage.js` & `searchRoutes.js`) features a multi-parameter filtering engine allowing users to locate exact items within seconds:

- **Full-Text Search:** Instant regex-based keyword search on item names with case-insensitivity (`$options: 'i'`).
- **Category Filtering:** Dynamic dropdown selection filtering items by designated rooms (Living, Bedroom, Bathroom, Kitchen, Office).
- **Condition Classification:** Direct filtering across physical condition tiers (New, Like New, Older).
- **Age Threshold Slider:** Interactive UI slider allowing users to constrain listings by maximum item age in years (using MongoDB `$lte` comparison).
- **Zero-State Feedback:** When no gifts match combined criteria, a clean, informative alert guides users to revise their filter parameters.

4.4 Product Details & Community Engagement

The details view (`DetailsPage.js`) provides deep insights for users evaluating a specific gift:

- **Expanded Information View:** Displays expansive item imagery, category designation, item condition, age in years, and detailed donor descriptions.
- **Authentication Barrier:** Guarantees that only authenticated community members can claim items or participate in comment discussions.
- **Community Comments Thread:** Chronological discussion thread where prospective recipients communicate with donors regarding pickup details and availability.
- **Intuitive Navigation:** Seamless browser history navigation allowing users to return to prior search results without state loss.

5. Artificial Intelligence & Sentiment Analysis Microservice

To maintain community trust and prevent harassment or spam within discussion threads, GiftLink integrates an autonomous Natural Language Processing (NLP) microservice running as an independent Node.js process.

5.1 Algorithmic Methodology

The sentiment engine utilizes Natural's Porter Stemmer algorithm combined with the AFINN lexical sentiment dictionary. The workflow comprises:

- **1. Tokenization & Stemming:** The incoming comment text is split into normalized token arrays and reduced to morphological base stems.
- **2. Valence Scoring:** The AFINN sentiment analyzer assesses the valence of individual words, calculating an aggregate sentiment score.
- **3. Categorical Thresholding:** The raw continuous score is classified into discrete sentiment states: Negative (< 0), Neutral (0 to 0.33), and Positive (> 0.33).
- **4. Microservice API Response:** The microservice responds with a JSON payload containing the exact float score and categorical tag, enabling real-time moderation.

```
// Sentiment Analysis Sample API Response
POST http://localhost:3000/sentiment?sentence=This%20gift%20is%20fantastic%20and%20helpful
HTTP/1.1 200 OK
Content-Type: application/json

{
  "sentimentScore": 0.6666666666666666,
  "sentiment": "positive"
}
```

6. Technology Stack & Framework Specifications

The technologies and libraries employed across the GiftLink platform are enumerated in the table below:

Layer / Domain	Library / Tool	Role & Engineering Utility
Frontend Client	React 18.2	Single Page Application architecture, functional components, hooks (useState, useEffect, useContext).
Routing & Nav	React Router DOM v6	Declarative client-side routing, protected routes, URL parameters (/app/product/:productId).
UI & Styling	Bootstrap 5 & CSS3	Responsive grid system, mobile-friendly navigation, customized cards, responsive tables.
Backend Server	Node.js & Express.js	Asynchronous event-driven REST API server managing routing, middleware, and database operations.
Database	MongoDB 6.x	NoSQL document store handling 'users' and 'gifts' collections with flexible schema support.
Database Driver	mongodb (Native Driver)	Asynchronous MongoClient connection pooling, query pipeline execution, and aggregation.
NLP & AI	natural (PorterStemmer/AFINN)	Tokenization, word stemming, and sentiment analysis for community moderation.
Authentication	jsonwebtoken (JWT)	Stateless, cryptographically signed user authorization tokens.
Security	bcryptjs	Adaptive salted password hashing protecting user credentials against brute-force attacks.
Observability	Pino & pino-http	High-performance structured JSON logging for request tracking and performance monitoring.
Config & Tools	dotenv, CORS, npm	Centralized environment configuration, Cross-Origin Resource Sharing control, and package management.

7. Database Schema & Data Modeling

GiftLink stores data within MongoDB in a dedicated database named `giftdb`. The schema utilizes two primary collections: `users` and `gifts`.

7.1 'users' Collection Schema

Field Name	Data Type	Description & Validation
_id	ObjectId	MongoDB auto-generated unique primary key.
email	String	User's unique email address (used as login identifier).
firstName	String	First name of the user.
lastName	String	Last name or surname of the user.
password	String	Salted and hashed representation of the user password (bcrypt).
createdAt	Date	Timestamp indicating user account registration date.

7.2 'gifts' Collection Schema

Field Name	Data Type	Description & Validation
id	String	External unique identifier for the gift listing (e.g., '875').
name	String	Title or item name of the gift (e.g., 'Dining Lamp').
category	String	Functional category ('Living', 'Bedroom', 'Kitchen', 'Office', 'Bathroom').
condition	String	Physical condition tier ('New', 'Like New', 'Older').
posted_by	String	Identifier or reference to the user who offered the gift.
zipcode	String	Postal zipcode indicating item location for local pick-up.
date_added	Number	Unix epoch timestamp of publication.
age_years	Number	Approximate age of the item in years (used in range filtering).
description	String	Narrative description highlighting item background and specifications.
image	String	Relative URL path to the item photograph asset.

8. RESTful API Specification & Endpoints

The core backend exposes a clean, standardized RESTful interface for all platform capabilities:

Method	Endpoint Path	Access	Operation Summary
POST	/api/auth/register	Public	Registers a new user account with hashed password and generates JWT token.
POST	/api/auth/login	Public	Authenticates user credentials and returns signed JWT authorization token.
PUT	/api/auth/update	Bearer JWT	Updates authenticated user profile metadata (e.g., display name).
GET	/api/gifts	Public	Retrieves complete array of all available gift listings from the database.
GET	/api/gifts/:id	Bearer JWT	Fetches detailed attributes of a specific gift by unique identifier.
POST	/api/gifts	Bearer JWT	Creates and publishes a new gift listing into the catalog.
GET	/api/search	Public	Executes multi-parameter query filtering by name, category, condition, and age.
POST	/sentiment	Public	Scores input sentences and returns sentiment valence and classification.

8.1 Sample Endpoint Payloads

Below are illustrative request/response contracts for core API operations:

```
// Registration & Login Endpoint Contract
POST /api/auth/register (or /api/auth/login)
Content-Type: application/json

Request Body:
{
  "email": "user@community.org",
  "password": "SecurePassword123!",
  "firstName": "Alex",
  "lastName": "Morgan"
}

Response (200 OK):
{
  "authtoken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
```

```
  "email": "user@community.org",
  "userName": "Alex"
}
```

9. Security Architecture, Quality Attributes & Observability

9.1 Defense-in-Depth Security

- **Cryptographic Hash Salting:** Plaintext passwords are never stored. Bcrypt applies salt rounds ensuring resistance to dictionary and rainbow-table attacks.
- **Stateless Token Verification:** Communications rely on signed JWT tokens containing claims expiration, preventing database lock contention on auth checks.
- **CORS Whitelisting:** Explicit Cross-Origin Resource Sharing (CORS) rules prevent unauthorized domain requests.
- **Environment Isolation:** Database connection strings and token secrets are loaded dynamically via dotenv, insulating secrets from source repositories.

9.2 High-Speed Observability & Logging

To satisfy production audit requirements, GiftLink embeds `pino` and `pino-http`. Unlike standard console loggers that block the Node.js event loop, Pino outputs serialized, high-throughput JSON telemetry streams. Every incoming HTTP request logs response status codes, execution latencies, client IPs, and payload volumes, enabling rapid operational diagnostics.

10. Containerization, Cloud Deployment & Scalability Strategy

GiftLink is architected from inception to adhere to the 12-Factor App methodology, facilitating containerized cloud deployment:

- **Docker Containerization:** Both frontend, backend, and sentiment services can be encapsulated in isolated Docker containers with pinned dependency runtimes.
- **Docker Compose Orchestration:** A unified Docker Compose specification coordinates local multi-container development, mounting MongoDB with volume persistence.
- **Elastic Horizontal Scaling:** Because services are stateless, backend replicas can scale horizontally behind an NGINX ingress controller or Kubernetes service mesh.
- **Cloud Platform Portability:** Native compatibility with IBM Cloud Code Engine, Red Hat OpenShift, AWS ECS, or Kubernetes clusters.

11. Future Product Roadmap & Conclusion

11.1 Future Roadmap Extensions

- **Real-Time Direct Messaging:** Introducing WebSocket / Socket.io based instant peer-to-peer chat for real-time pickup coordination.
- **Interactive Geolocation Radius Map:** Interactive OpenStreetMap / Mapbox integration allowing neighbors to discover gifts within a configurable radius.
- **Reputation & Karma Engine:** User ratings, donor badges, and community trust karma points rewarding active contributors.
- **Push & Email Alerts:** Automated mobile and desktop notifications when new gifts matching saved filter queries are posted.

11.2 Conclusion

The GiftLink Capstone Project delivers a complete, cohesive, and socially transformative software solution. By merging modern responsive single-page web engineering, decoupled microservice design patterns, robust NoSQL document storage, and artificial intelligence for community sentiment evaluation, GiftLink stands as an exemplary demonstration of full stack software craftsmanship.

12. Capstone Submission Verification & Evaluation Details

> **SUBMISSION VERIFICATION CHECKLIST**
> *Submission Email Target: meharabrehman@gmail.com*
> *Submission Format: PDF (Converted from this formatted DOCX document)*
> *Subject / Body Requirement: Includes Student Names and Project Title*
> *Evaluator Notice: Complete, production-ready system specification ready for grading.*

Project Title	GiftLink - Full Stack Community Resource Sharing Platform
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Artifact Deliverable	Formatted Capstone Idea & System Specification Document (DOCX ➔ PDF)
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